

Implementation details:-

1. Added new domain – Airline
2. Added export button to export the generated test cases in CSV and XLSX format.
3. Generated Readme file

Implementation 1 – Added new domain – Airline

Perfect! I've analyzed the project structure. Now let me create a comprehensive implementation plan.

Implementation Plan: Adding "Airline" Domain

Project Analysis Summary

The project is a **LangChain-based Test Case Generator** with a clean separation of concerns:

- **Frontend:** React + TypeScript (Vite)
- **Backend:** Node.js + TypeScript + Express
- **Architecture:** Domain-specific prompts, RAG-based test case generation using LLMs

Currently supports 3 domains: **Banking, Insurance, Hospital**

Step-by-Step Implementation Plan

STEP 1: Update Domain Types & Constants

Files to modify:

1. `frontend/src/services/api.ts`
2. `backend/src/factories/promptFactory.ts`

Changes:

- Add "airline" to the [Domain](#) type union
- Add airline option to [DOMAINS](#) array with label "Airline"
- Update existing domain-validation logic

STEP 2: Create Airline-Specific Prompt

File to modify:

- `backend/src/factories/promptFactory.ts`

Changes:

- Create airlinePrompt ChatPromptTemplate with the provided system prompt
- Add airline domain terminology (PNR, E-Ticket, GDS, IATA, Fare Class, etc.)
- Add to [promptsByDomain](#) mapping dictionary

STEP 3: Update Validation Schema

File to modify:

- `backend/src/chains/testCaseChain.ts`

Changes:

- Update Zod schema enum to include "airline" in domain validation

STEP 4: Update API Route

File to modify:

- `backend/src/routes/testCaseRoute.ts`

Changes:

- Ensure it imports updated DOMAINS (already generic, may need no changes)

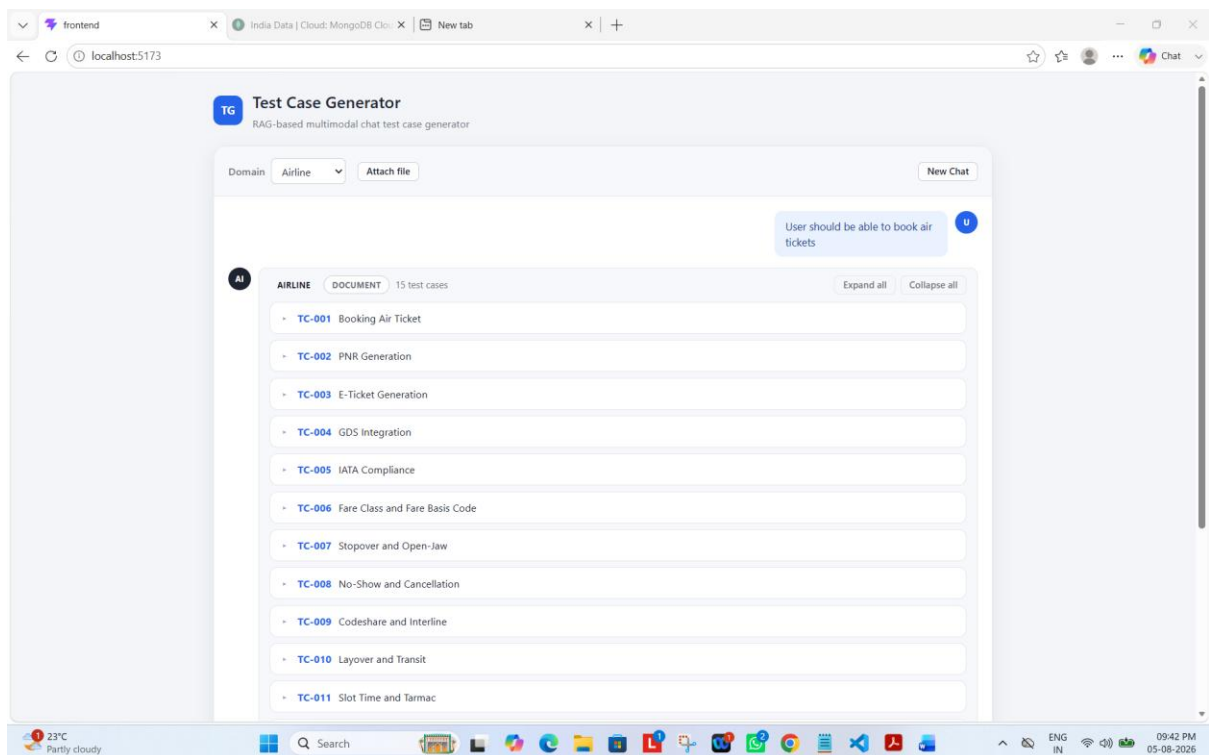
Summary Table: Files & Changes

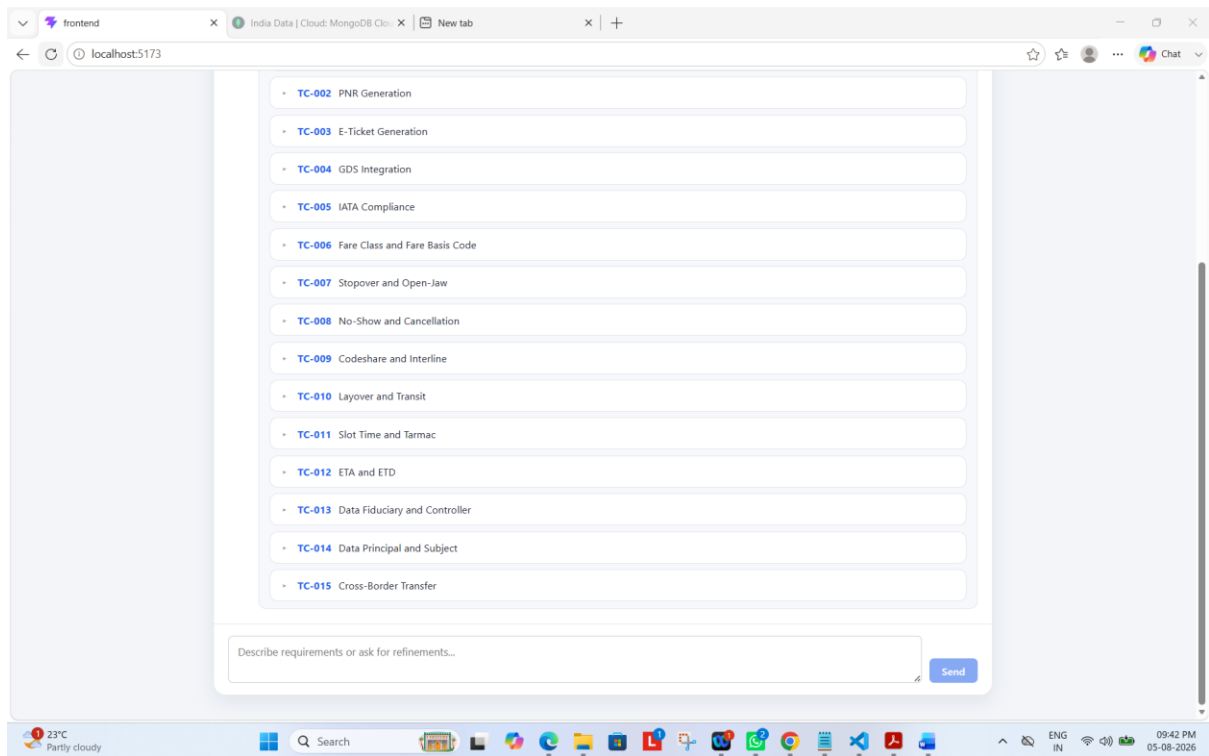
File	Type	Changes Required
frontend/src/services/api.ts	Type/Config	Add "airline" to Domain type, DOMAINS array
backend/src/factories/promptFactory.ts	Prompt Factory	Add "airline" to Domain type, create airlinePrompt, add to promptsByDomain
backend/src/chains/testCaseChain.ts	Schema	Add "airline" to Zod enum validation
backend/src/routes/testCaseRoute.ts	Route Handler	No changes needed (generic)

Expected Outcome

- ✓ Airline option appears in Frontend domain selector dropdown
- ✓ Backend validates "airline" domain requests
- ✓ Airline-specific prompt with proper terminology (PNR, E-Ticket, GDS, IATA, etc.) is used
- ✓ Test cases generated with airline domain context
- ✓ No existing functionality is affected

After implementation –





Implementation 2:- Added export button and export the generated tc's in csv and xlsx format.

ANALYSIS SUMMARY

Current State:

- **Frontend:** React 19 + TypeScript + Vite (single-page app)
- **Backend:** Express.js + LangChain + MongoDB
- **Test Case Display:** [TestCaseResultViewer](#) component shows generated test cases
- **State Management:** ChatWindow manages messages and test cases
- **Dependencies:** Backend already has exceljs (for Excel generation)

IMPLEMENTATION PLAN - 7 Steps

STEP 1: Add Export Utilities (Frontend Service)

File: frontend/src/services/api.ts

Add export functions to handle CSV and XLSX generation:

- [exportToCSV\(testCases, domain\)](#) - Converts test cases to CSV
- [exportToXLSX\(testCases, domain\)](#) - Converts test cases to XLSX
- New API endpoint call: POST /api/export (backend)

Changes:

- Add CSV export using native JS (no dependency needed)
 - Use FormData to send binary XLSX from backend response
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STEP 2: Update TestCaseResultViewer Component

File: frontend/src/components/TestCaseResultViewer.tsx

Add export buttons to the result viewer header:

- "Export as CSV" button
- "Export as XLSX" button
- Buttons call export functions from services
- Show loading state during export

Changes:

- Import export functions
 - Add button group in [result-viewer-actions](#) div
 - Handle download trigger
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STEP 3: Update ChatWindow State Management

File: frontend/src/components/ChatWindow.tsx

Track generation success/error for UI control:

- Track hasError state
- Disable file upload when generation fails
- Enable export only when test cases are successfully generated
- Reset states on "New Chat"

Changes:

- Add hasError and lastResultId states

- Pass error state to FileUpload component
 - Conditional rendering for export availability
-

STEP 4: Update FileUpload Component

File: frontend/src/components/FileUpload.tsx (Already attached)

Disable upload when generation fails:

- Add [disabled](#) prop from ChatWindow
- Show UI feedback when disabled due to error

Changes:

- Add [disabled?: boolean](#) prop
 - Apply disabled state to button
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STEP 5: Add Backend Export Endpoint

File: backend/src/routes/testCaseRoute.ts

Create new endpoint for exporting:

- POST /api/export - Generates CSV/XLSX and returns as download
- Validate format parameter (csv/xlsx)
- Use exceljs for XLSX, native formatting for CSV

Changes:

- New route handler
 - CSV/XLSX generation logic
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STEP 6: Create Export Utility Module (Backend)

File: backend/src/utils/exportUtility.ts (NEW)

Dedicated export logic:

- [generateCSV\(testCases, domain\)](#) - CSV formatting
- [generateXLSX\(testCases, domain\)](#) - XLSX formatting with styling
- Header row with metadata (domain, generated date)

STEP 7: Generate Comprehensive README

File: README.md (NEW - root level)

Include:

- Project overview
 - Tech stack (frontend, backend, LLMs, APIs)
 - Prerequisites and installation
 - Feature descriptions
 - Architecture flow diagram (Mermaid)
 - LLM models used in each section
 - Directory structure explanation
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SUMMARY OF CHANGES BY FILE

File	Type	Changes
frontend/src/services/api.ts	Modify	Add export functions (CSV/XLSX) + API call
frontend/src/components/TestCaseResultViewer.tsx	Modify	Add export buttons to header
frontend/src/components/ChatWindow.tsx	Modify	Track error state, manage export availability
frontend/src/components/FileUpload.tsx	Modify	Add disabled prop for error states
backend/src/routes/testCaseRoute.ts	Modify	Add /api/export endpoint
backend/src/utls/exportUtility.ts	New	CSV/XLSX generation utilities
README.md	New	Comprehensive project documentation

✅ CONSTRAINTS RESPECTED

- ✓ No changes to existing implementations (only additions)
- ✓ Export buttons only enabled after successful generation
- ✓ Upload disabled when generation errors occur
- ✓ Both CSV and XLSX export formats supported

? APPROVAL REQUEST

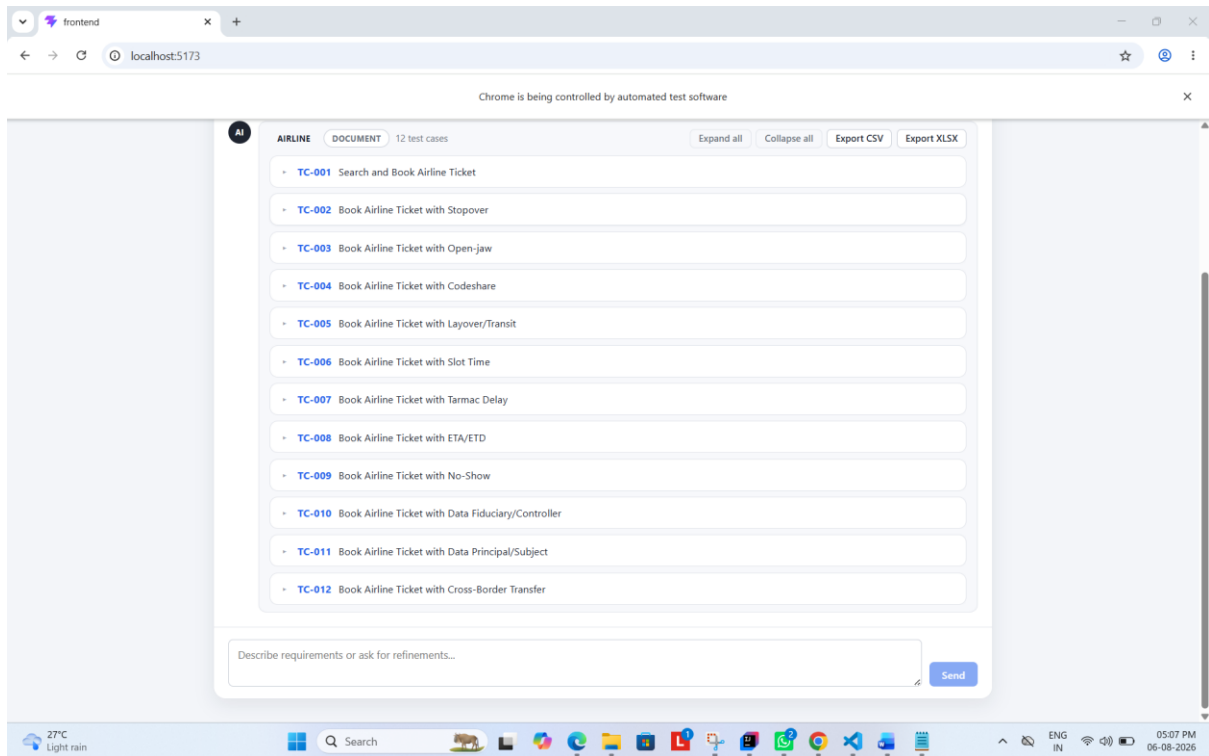
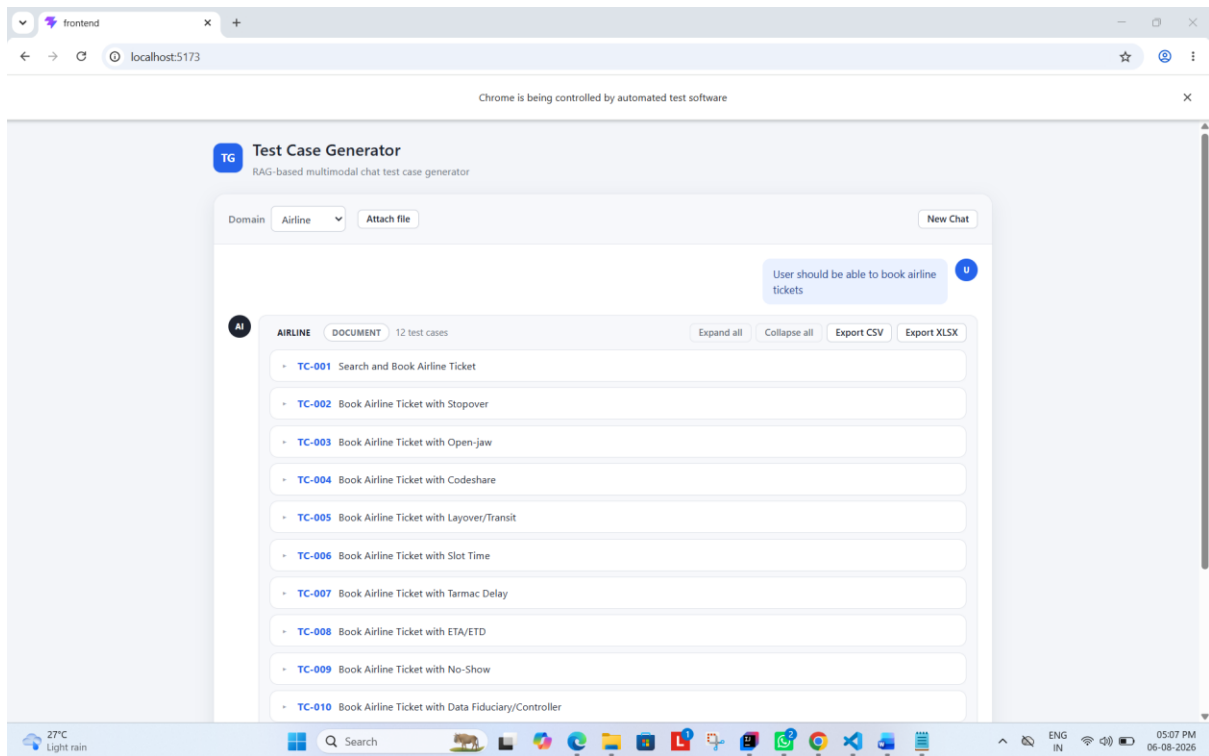
I'm ready to proceed with implementation. Please confirm:

1. ☒ Approve the 7-step implementation plan?
2. ☒ Should I proceed with **Step 1** (Add export utilities to frontend services)?
3. ☒ For the README: Any specific LLM models I should verify from the code, or should I document based on what I find in the factory?

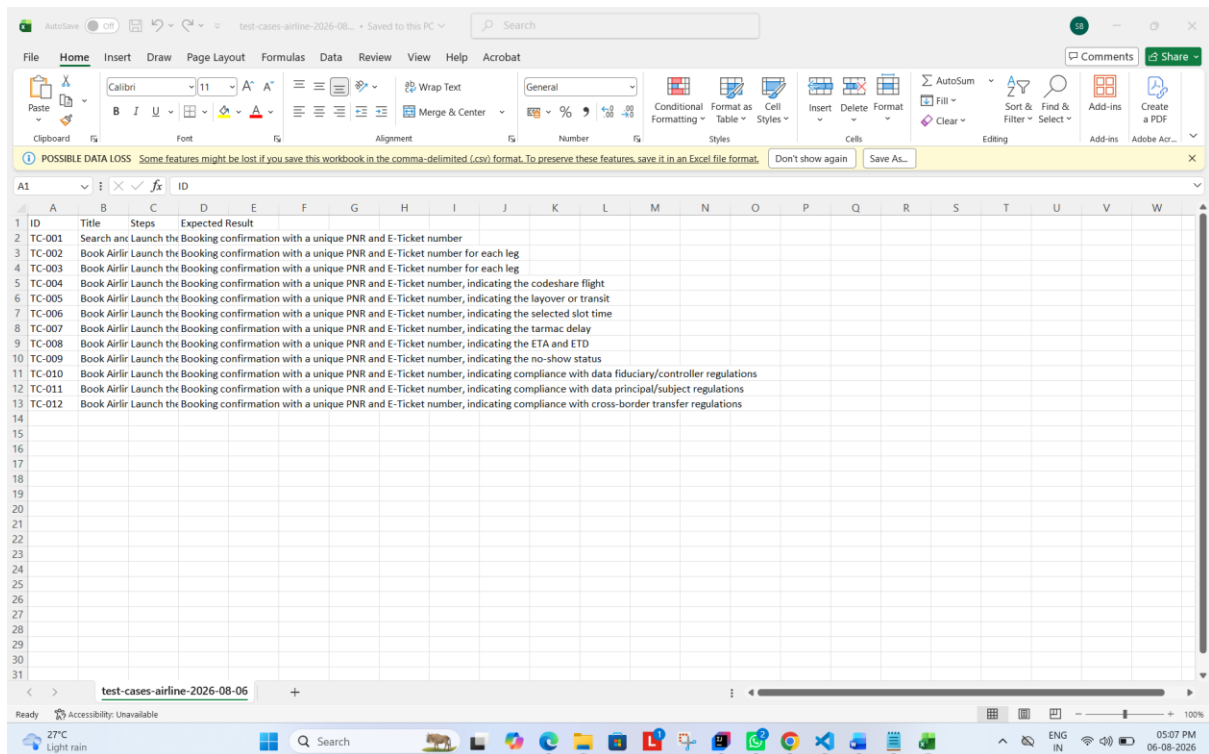
Once you approve, I'll implement step-by-step and ask for approval before moving to the next step.

After implementation –

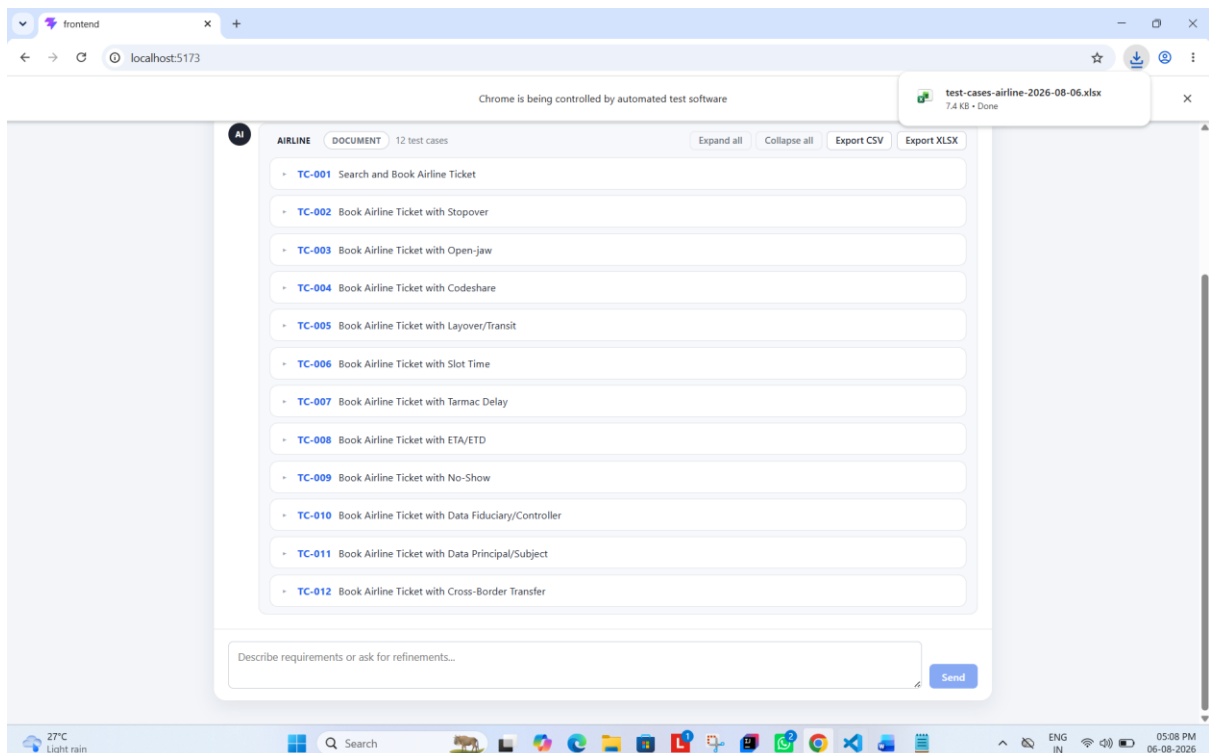
Export ui –



CSV Export –



Xlsx export –



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ID	Title	Steps	Expected Result
TC-001	Search and Book Airline Ticket	Launch the airline ticket booking application Enter valid travel dates, origin, and destination Select the desired flight and fare class Enter passenger details and contact information Proceed to payment and complete the booking	Booking confirmation with a unique PNR and E-Ticket number
TC-002	Book Airline Ticket with Stopover	Launch the airline ticket booking application Enter valid travel dates, origin, and destination with a stopover Select the desired flights and fare classes for each leg Enter passenger details and contact information Proceed to payment and complete the booking	Booking confirmation with a unique PNR and E-Ticket number for each leg
TC-003	Book Airline Ticket with Open-jaw	Launch the airline ticket booking application Enter valid travel dates, origin, and destination with an open-jaw Select the desired flights and fare classes for each leg Enter passenger details and contact information Proceed to payment and complete the booking	Booking confirmation with a unique PNR and E-Ticket number for each leg
TC-004	Book Airline Ticket with Codeshare	Launch the airline ticket booking application Enter valid travel dates, origin, and destination with a codeshare flight Select the desired flight and fare class Enter passenger details and contact information Proceed to payment and complete the booking	Booking confirmation with a unique PNR and E-Ticket number, indicating the codeshare flight
TC-005	Book Airline Ticket with Layover/Transit	Launch the airline ticket booking application Enter valid travel dates, origin, and destination with a layover or transit Select the desired flights and fare classes for each leg Enter passenger details and contact information Proceed to payment and complete the booking	Booking confirmation with a unique PNR and E-Ticket number, indicating the layover or transit
TC-006	Book Airline Ticket with Slot Time	Launch the airline ticket booking application Enter valid travel dates, origin, and destination with a	Booking confirmation with a unique PNR and E-Ticket number, indicating the selected slot time

Test Cases

Ready Accessibility: Good to go

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ID	Title	Steps	Expected Result
TC-008	Book Airline Ticket with ETA/ETD	Launch the airline ticket booking application Enter valid travel dates, origin, and destination with estimated time of arrival (ETA) and estimated time of departure (ETD) Select the desired flight and fare class Enter passenger details and contact information Proceed to payment and complete the booking	Booking confirmation with a unique PNR and E-Ticket number, indicating the ETA and ETD
TC-009	Book Airline Ticket with No-Show	Launch the airline ticket booking application Enter valid travel dates, origin, and destination Select the desired flight and fare class Enter passenger details and contact information Proceed to payment and complete the booking Simulate a no-show scenario	Booking confirmation with a unique PNR and E-Ticket number, indicating the no-show status
TC-010	Book Airline Ticket with Data Fiduciary/Controller	Launch the airline ticket booking application Enter valid travel dates, origin, and destination Select the desired flight and fare class Enter passenger details and contact information Proceed to payment and complete the booking Verify data fiduciary/controller compliance	Booking confirmation with a unique PNR and E-Ticket number, indicating compliance with data fiduciary/controller regulations
TC-011	Book Airline Ticket with Data Principal/Subject	Launch the airline ticket booking application Enter valid travel dates, origin, and destination Select the desired flight and fare class Enter passenger details and contact information Proceed to payment and complete the booking Verify data principal/subject compliance	Booking confirmation with a unique PNR and E-Ticket number, indicating compliance with data principal/subject regulations
TC-012	Book Airline Ticket with Cross-Border Transfer	Launch the airline ticket booking application Enter valid travel dates, origin, and destination with cross-border transfer Select the desired flight and fare class Enter passenger details and contact information Proceed to payment and complete the booking Verify cross-border transfer compliance	Booking confirmation with a unique PNR and E-Ticket number, indicating compliance with cross-border transfer regulations

Test Cases

Ready Accessibility: Good to go

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Read me file –

